Benjamin Alt

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Professional Experience

AICOR Solutions
Bremen, Germany

Co-Founder and Research Lead *Apr 2025 - present*

- Co-develop system architectures for virtual commissioning and adaptive mobile manipulation
- Design transfer pipelines for integrating research-grade robot learning methods into industrial applications

AICOR Institute for Artificial Intelligence, University of Bremen
Bremen, Germany

Technical Director *Apr 2025 - Apr 2026*

- Led the development of a software ecosystem for open-source cognitive robotics
- Co-led research on hybrid AI for robot perception and manipulation
- Translated research prototypes into deployable systems in collaboration with spin-off and industrial partners
- Acquired third-party funding for research and technology transfer

ArtiMinds Robotics
Karlsruhe, Germany

Senior Team Lead Research *Oct 2024 - Mar 2025*

- Supervised a team of seven researchers developing machine learning methods for robot manipulation and control
- Directed integration of AI-based perception and control models into commercial robotic programming tools
- Led 8 publicly funded research projects on cognitive robotics for industrial manipulation
- Established and expanding long-term research partnerships with >20 academic institutions and >15 industry partners

Senior Research Scientist *Jan 2023 - Sep 2024*

- Researched and developed methods for AI-enabled robot programming and manipulation (8 conference papers)
- Acquired and realized 5 publicly funded research projects
- Provided technical leadership on applying AI methods to industrial automation challenges
- Mentored and supervised 14 graduate and undergraduate students

Research Scientist *Oct 2019 - Dec 2022*

- Researched and published on semi-symbolic robot program inference with deep neural networks (5 conference papers, 2 book chapters)
- Implemented and patented a commercial AI solution for the data-driven optimization of industrial production processes
- Acquired and realized 6 publicly funded research projects
- Mentored and supervised 16 graduate and undergraduate students

Education

University of Bremen
Bremen, Germany

Ph.D. Computer Science *magna cum laude* *2020 - 2025*

- Dissertation: *Neurosymbolic Robot Programming - A Framework for AI-Enabled Programming of Robot Manipulation Tasks*
- Advisor: Prof. Michael Beetz, AICOR Institute for Artificial Intelligence

Karlsruhe Institute of Technology
Karlsruhe, Germany

M.Sc. Computer Science *with distinction* *2017 – 2019*

- Thesis: *Automatic Parameterization of Robot Programs via Learning of Neural Program Representations*
- Merit scholarship of the German Acad. Scholarship Foundation (Studienstiftung des deutschen Volkes)

B.Sc. Computer Science *2015 – 2017*

- Thesis: *Machine Learning for Pose Optimization: An Integrated Framework for the Development and Monitoring of Adaptive Robot Programs*

Institut d’Études Politiques de Paris
Reims, France

B.A. Political Science *summa cum laude* *2012 – 2015*

Selected Publications

- B. Alt, C. Kienle, D. Katic, R. Jäkel, and M. Beetz, "Shadow Program Inversion with Differentiable Planning: A Framework for Unified Robot Program Parameter and Trajectory Optimization", in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2025. DOI: `10.48550/arXiv.2409.08678`
- B. Alt et al., "Domain-Specific Fine-Tuning of Large Language Models for Interactive Robot Programming", in *European Robotics Forum 2024*, C. Secchi and L. Marconi, Eds., vol. 32, Springer Nature Switzerland, 2024, pp. 274–279. DOI: `10.1007/978-3-031-76424-0_49`
- B. Alt et al., "RoboGrind: Intuitive and Interactive Surface Treatment with Industrial Robots", in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 1–8. DOI: `10.1109/ICRA57147.2024.10611143`
- B. Alt, D. Katic, R. Jäkel, and M. Beetz, "Heuristic-Free Optimization of Force-Controlled Robot Search Strategies in Stochastic Environments", in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 8887–8893. DOI: `10.1109/IROS47612.2022.9982093`
- B. Alt, D. Katic, R. Jäkel, A. K. Bozcuoglu, and M. Beetz, "Robot Program Parameter Inference via Differentiable Shadow Program Inversion", in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2021, pp. 4672–4678. DOI: `10.1109/ICRA48506.2021.9561206`

Full list of publications: benjaminalt.github.io/publications

Skills

Robotics	Task and motion planning, force control, 3D visual perception, robot programming, human-robot interaction, model predictive control, manipulation of deformables
Machine learning	Deep learning, imitation learning, learning from demonstration, differentiable programming, model-based optimization, interpretability, informed machine learning
Research management	Grant acquisition, science communication, stakeholder management, technology transfer, strategic planning
Leadership	Team leadership, mentoring, talent acquisition
Programming languages	Python (9 years of professional experience), C++ (3 years)
Development tools	Git, Cursor, DVC, Jira, CMake, Jenkins CI
Frameworks	PyTorch, NumPy, Open3D, MoveIt, Keras, ROS, Qt

Karlsruhe, August 8, 2026

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